A Language Model from 1913: Pretraining on Historical Text

Xiaoxi Luo[1,2], Zachary Shinnick[3,4], Niclas Griesshaber[4,5], Yixuan Wang[1,2], Junchi Yu[4], Freda Shi[1,2], Philip Torr[4], Yao Lu[6]

[1]University of Waterloo [2]Vector Institute [3]Adelaide University
[4]University of Oxford [5]Oxford Centre for Economic and Social History
[6]University College London

EMNLP2026 BUDAPEST

Query

TypewriterLM Response

Do you think there will soon be a major war in Europe?

"Certainly... It would not surprise me if this country were to go to war with France within ninety days."

Do you know about the general theory of relativity by Albert Einstein?

"...Professor A. Einstein, of whom we have not yet learned to speak as a physicist..."

What is a computer?

"...an operator who multiplies or divides... a person who performs arithmetical operations..."

Teaser. Probing TypewriterLM on "future" events.

Takeaways

- **54B historical tokens** produce a capable 7.24B model with a **1913 cutoff**.
- **Leakage-aware post-training** constrains every assistant response to pre-1913 source language.
- **The cutoff is visible:** History LMs become more surprised by post-cutoff events.
- **Capability remains non-trivial:** TypewriterLM is competitive with existing History LMs on historically aligned benchmarks.
- **Metadata leakage is small but real** and requires document-level filtering.

History LMs must be evaluated for capability and cutoff adherence together.

1. Pretraining

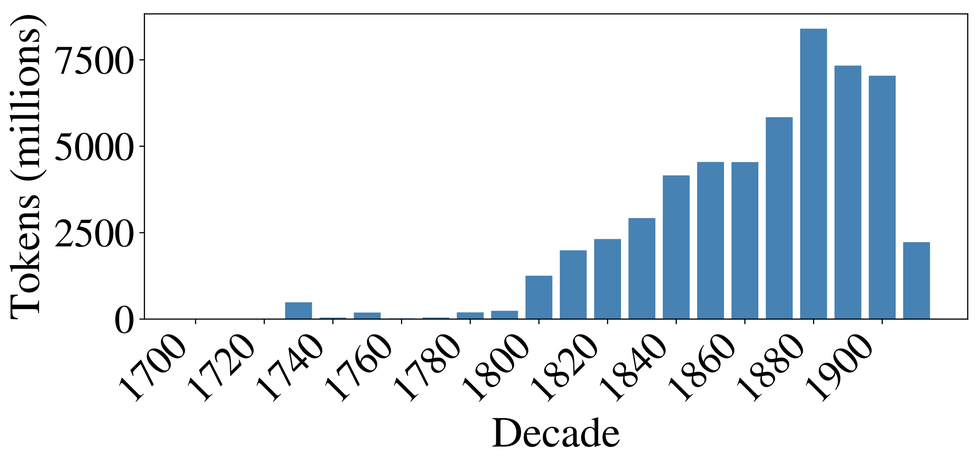


Figure 1. TypewriterCorpus tokens by decade.

54B corpus tokens **140B training tokens** **2.59 epochs**

Model

- Llama 3-style decoder, 32 layers
- 7.24B parameters; 8,192 context
- 32K custom historical BPE
- GQA: 32 query / 8 KV heads

Leakage-aware cleaning

- OCR-quality and fragment filters
- Repair split words and OCR artifacts
- Remove provenance, URLs, HTML, and editorial metadata

Dataset	Domain	Tok. (M)	%
Institutional	Books	52,736	97.68
BL Books	Books	915	1.69
Hansard	Parliamentary	193	0.36
Royal Society	Scientific	70	0.13
Old Bailey	Court	39	0.07
CLMET	Literary	31	0.06
CED	Dialogues	0.67	<0.01
Lampeter	Tracts	0.35	<0.01
CLME Prose	Prose	0.12	<0.01
Pamphlets	Pamphlets	0.08	<0.01
Total		**53,986**	**100**

Table 1. Corpus composition after filtering.

2. Post-training

Pre-1913 source passage p → Historical response y → Lexical verification → Generated instruction

Strict grounding: $V(y) \subseteq V(p) \cup A$; every content word and numeral must be supported by the historical passage.

1,030
History-LIMA
Human-reviewed, including 30 multi-turn dialogues; 15 epochs.

287,860
History-SelfInstruct
Responses stay fixed; only instructions are generated; 1 epoch.

3. Evaluation I: Knowledge Cutoff Adherence

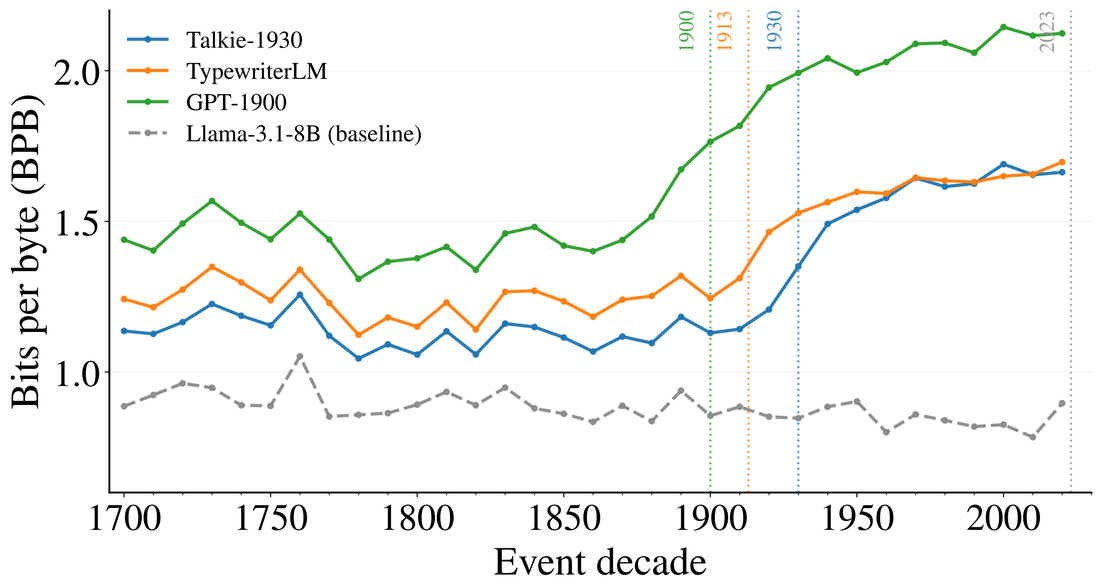
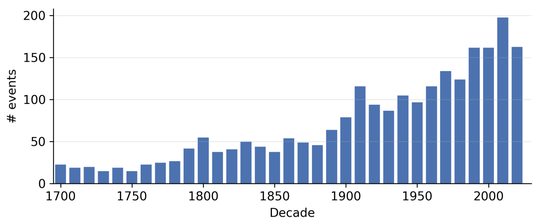


Figure 2. BPB surprisingness rises around History LM cutoffs.



History-Event: 2,344 events spanning 1700–2025.

Two complementary tests

1. **Surprisingness:** bits per byte for event description and year.
2. **Recall:** strict and relaxed factual recall before and after cutoff.

1,744 validated recall questions remain after filtering.

			Correctness (pre-cutoff)↑		Leakage (post-cutoff)↓		Count	
Model	**Size**	**Cutoff**	**Strict**	**Relaxed**	**Strict**	**Relaxed**	**pre**	**post**
Talkie-1930 (base)	13B	1930	2.9	8.8	0.0	0.3	719	1025
Talkie-1930 (it)	13B	1930	34.5	53.3	0.0	0.9	719	1025
GPT-1900 (sft)	3.29B	1900	5.6	13.5	0.0	0.2	498	1246
Mr. Chatterbox	340M	1899	0.2	1.2	0.0	0.1	492	1252
TimeCapsuleLLM	1.2B	1875	0.0	0.5	0.0	0.0	394	1350
TypewriterLM (base)	7.24B	1913	0.9	6.1	0.0	0.4	576	1168
TypewriterLM (Self-Instruct)	7.24B	1913	**6.2**	20.3	0.1	0.6	576	1168
TypewriterLM (LIMA)	7.24B	1913	3.6	**20.5**	0.0	**0.3**	576	1168

Table 2. Full factual correctness and post-cutoff leakage results on History-Event.

4. Evaluation II: Benchmarks

General capability

Model	Era	Size	ARC-E	ARC-C	HellaSwag
Mr. Chatterbox	1837–1899	0.34B	33.9	22.9	28.2
TimeCapsule-v2	1800–1875	1.22B	30.3	22.0	25.9
GPT-1900	pre-1900	3.29B	36.7	23.9	32.3
Talkie-1930	pre-1930	13B	55.8	35.6	49.8
GPT-2 XL	pre-2019	1.55B	50.8	28.2	48.8
Qwen3-4B	pre-2025	4B	77.4	51.6	65.9
TypewriterLM	**1700–1913**	**7.24B**	**43.3**	**27.9**	**36.1**

Table 3. Accuracy (%) on ARC-Easy, ARC-Challenge, and HellaSwag.

TypewriterLM outperforms smaller History LMs and reaches GPT-2 XL-level ARC-Challenge accuracy despite its 1913 cutoff.

Temporal mismatch changes HellaSwag

Model	Original	Filtered	1800s prose
GPT-2 XL	48.8	48.2	41.2
Qwen3-4B	65.9	66.2	54.0
GPT-1900	32.3	32.3	36.0
Talkie-1930	49.8	51.2	52.3
TypewriterLM	**36.1**	**36.2**	**40.0**

Table 4. Accuracy (%) after topic filtering and historical-style rewriting.

Rewriting into 1800s prose improves History LMs while reducing modern-model accuracy: language style is a major evaluation confound.

Historically filtered AlpacaEval

Model vs. GPT-1900	Win	Loss
TypewriterLM (LIMA)	46.5	53.5
TypewriterLM (Self-Instruct)	42.0	58.0

Table 5. Pairwise rates (%) on 376 historically valid prompts.

LIMA remains competitive with GPT-1900 even though every response is historically grounded.

Historically filtered IFEval

Model	P-S	P-L	I-S	I-L
GPT-1900	6.8	7.2	15.4	16.9
Talkie-1930-IT	11.8	11.8	22.6	22.6
TypewriterLM (LIMA)	4.6	5.3	13.5	14.2
TypewriterLM (Self-Instruct)	10.3	10.3	18.1	19.1

Table 6. Prompt/instruction accuracy under strict/loose matching.

Self-Instruct broadly matches existing History LMs without unrestricted modern response generation.

Overall: historically aligned evaluation changes how capability should be interpreted.